Electric Vehicle

Category: **Automotive** Type: **Full SOP** Priority: **High** 5 labs

Covers these lab items: Four wheeler Electric vehicle model; Four wheelers old second hand Electric vehicle; Old second hand Electric two-wheeler

What it is used for

An electric vehicle uses a traction battery, electric motor, and power electronics to provide vehicle movement. Safe inspection, correct charging practices, and approved operating procedures help protect occupants and technicians while maintaining vehicle performance.

Main parts

- Traction battery stores electrical energy for propulsion.
- Electric motor converts electrical energy into mechanical movement.
- Inverter or motor controller manages power supplied to the motor.
- Vehicle control unit coordinates vehicle systems and operating functions.
- Charging port connects the vehicle to compatible charging equipment.
- Onboard charger converts AC charging power when supported by the vehicle.
- Battery management system monitors battery conditions and protection limits.
- Braking system, tyres, lights, and safety restraints support safe driving.

Before you start

1. Confirm the operator has the required driving licence and vehicle authorisation.
2. Read the vehicle manual and identify its charging type and operating instructions.
3. Inspect tyres, brakes, lights, mirrors, windscreen, and visible body damage.
4. Check the dashboard for warning lights and verify the battery state of charge.
5. Inspect the charging cable and connector for damage before charging.
6. Ensure the vehicle has no visible fluid leaks, exposed high-voltage wiring, or unusual damage.

How to use

1. Enter the vehicle and adjust the seat, mirrors, and steering position safely.
2. Fasten the seat belt and ensure all occupants use appropriate restraints.
3. Check the surroundings and start or activate the vehicle using the prescribed procedure.
4. Select the required drive mode and confirm the dashboard shows readiness to drive.
5. Release the parking brake as instructed and move off smoothly.
6. Follow traffic rules, speed limits, and safe driving distances throughout the journey.
7. Monitor battery level, warning indicators, and vehicle response while driving.
8. Park in a suitable location, apply the parking brake, and switch off the vehicle.
9. For charging, use compatible approved equipment and follow the vehicle's charging sequence.
10. After charging, stop the session as instructed, disconnect the connector safely, and secure the charging port.

⚠ Safety

- Never touch exposed high-voltage cables, connectors, or damaged battery components.
- Do not operate the vehicle if critical brake, steering, tyre, or high-voltage warnings appear.
- Use only compatible charging equipment and avoid charging with damaged cables or connectors.
- Keep charging connectors dry and follow the manufacturer's instructions for wet conditions.
- Do not open the traction battery or attempt high-voltage repairs without specialised training and authorisation.
- If smoke, unusual heat, strong chemical odour, or battery damage appears, stop safely, move away, and contact emergency or authorised service personnel.

Care & storage

- Check tyre pressure, tread, brakes, lights, and washer fluid at recommended intervals.
- Keep the charging port and connector clean, dry, and free from debris.
- Follow the manufacturer's battery charging, parking, and maintenance recommendations.
- Arrange inspection by authorised service personnel when warnings, reduced performance, or unusual noises occur.

Common mistakes to avoid

- Driving without checking battery charge, tyres, brakes, or dashboard warnings.
- Using an incompatible or damaged charging cable.
- Ignoring high-voltage warnings or attempting unauthorised battery repairs.
- Leaving the vehicle in an unsuitable charging or parking condition contrary to the manual.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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